Reimplementation of LoRA: Low-Rank Adaptation of Large Language Models

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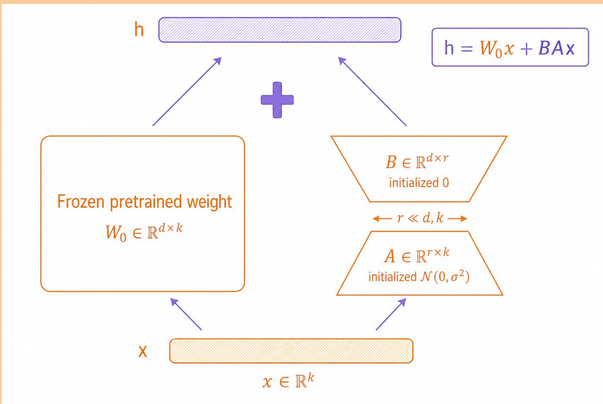
LoRA Background

- Motivation: How can we make fine-tuning parameter efficient?
- LoRA: Learn a low-rank update to the original model weights

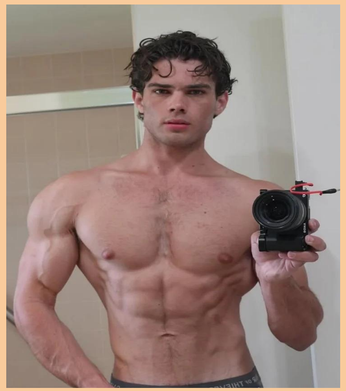


- Learn A and B with a low rank (r)
- These matrices serve as the update to the original model
- Very few trainable parameters
- BA has the same shape as the targeted layer weight matrix W_0

Our Methodology:

- Original Model: RoBERTa-base (~125M parameters)
- Dataset/Task: GLUE SST-2 (sentiment analysis of movie reviews)
- Baseline Comparison: Full Fine-Tuning vs. LoRA (r = 8, target modules = Q + V)

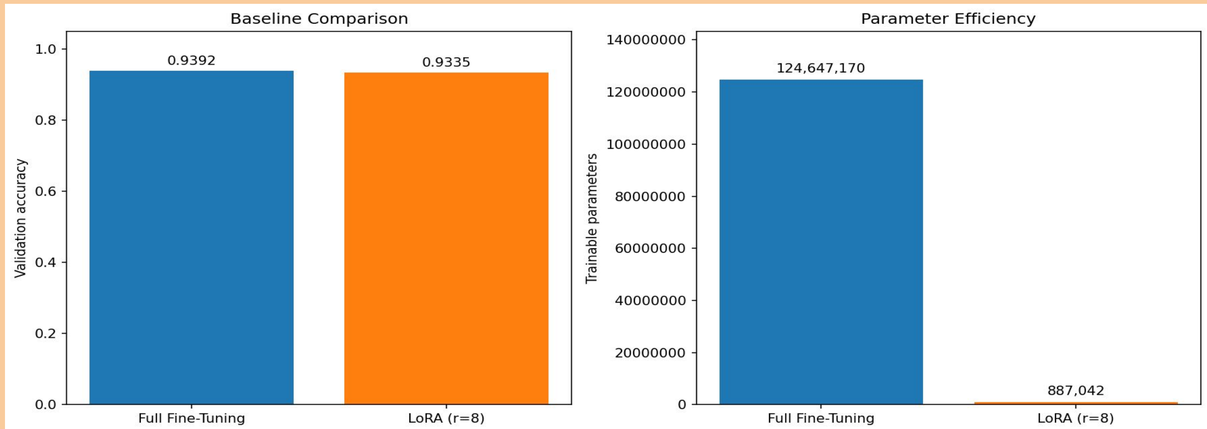
Big Bloated Full Fine-Tuning



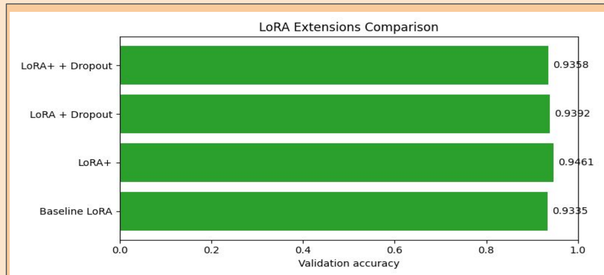
Slim and Efficient LoRA Fine-Tuning

Full Fine-Tuning Brutally Parameter Mogged By LoRA

Key Question: Can LoRA match full fine-tuning with <1% of the trainable parameters?



Accuracy stays within 0.57 percentage points of full fine-tuning while trainable parameters drop by ~140x

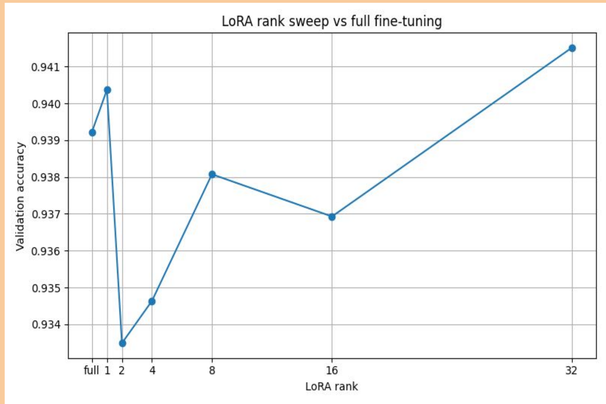


LoRA Variant Experimentation

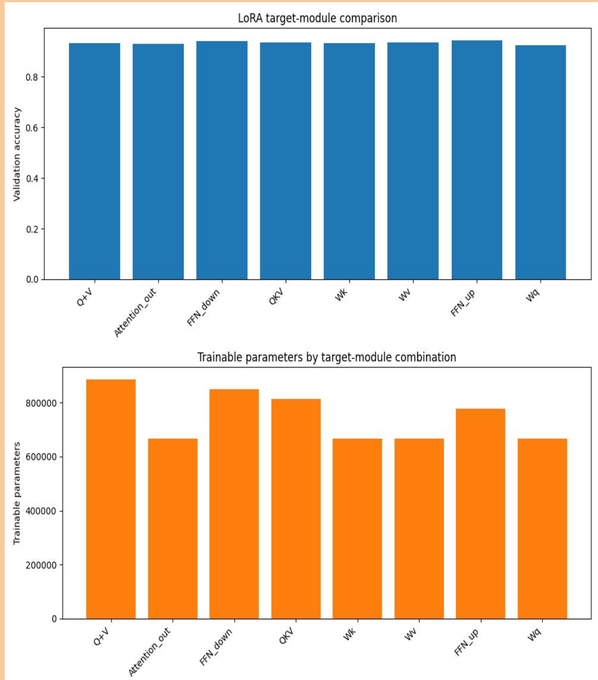
LoRA+ was the strongest variant at ~94.6% validation accuracy

*Note: each experiment uses different number of epochs, so results are relevant only for comparisons within experiments

Hyperparameter Experimentation



Low ranks are viable for SST-2, suggesting the task may have a low intrinsic adaptation dimension



Several target modules were competitive on SST-2, with FFN_up performing best

Potential Future Work: study generalization after fine-tuning and test tasks with higher intrinsic dimensionality

References: Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685; Hayou, S., Ghosh, N., and Yu, B. (2024). LoRA+: Efficient Low Rank Adaptation of Large Models. arXiv:2402.12354.